

Session 5 · Deck 5.A1

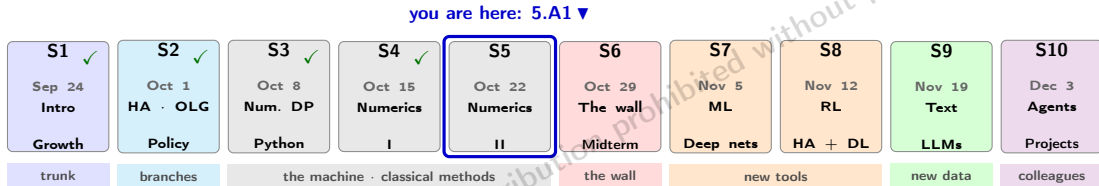
What Perturbation Computes

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Thursday, October 22, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 5.A1 what perturbation computes · 5.A2 the model and the parameter · 5.A3 first order · 5.A4 linear RE systems · 5.A5 higher order, risk, pruning · 5.A6 the worked example.

Hook: Four Numbers Against Two Thousand

Session 4 finished by solving the quarterly RBC model as well as a grid can: continuous choice, linear interpolation, golden section, Howard. It stores **2,100 numbers** — one per (k, z) — and takes about a second.

By the end of this block we will solve the same model with **four numbers**, in well under a millisecond, and near the steady state it will be **more accurate**.

That should be alarming. Understanding exactly why it happens — and exactly where it stops being true — is the whole of Block A.

Ledger three_methods. "Near" is $k \in [0.95k^*, 1.05k^*]$, "wide" is $[0.5k^*, 1.5k^*]$; $\mathcal{E}$ is the unit-free Euler residual of 3.A1.

	grid (4.B5)	1st order
numbers stored	2,100	4
solve time	0.94 s	< 0.001 s
$\max \mathcal{E} $ near k^*	$10^{-2.82}$	$10^{-4.01}$
$\max \mathcal{E} $ wide band	$10^{-2.51}$	$10^{-2.56}$

Read the last row before celebrating. Away from the steady state the advantage is gone.

What We Are Actually Solving

Every model in this course reduces to one line: find an unknown **function** d — a decision rule — with

$$H(d, x, \theta) = 0,$$

where x is the state and θ the parameters.

Sessions 3 and 4 stored d at N grid points and iterated.

That is one way to represent a function.

Perturbation uses a **series around a point where the answer is known**,

$$d^n(x, \theta) = \sum_{i=0}^n a_i (x - x_0)^i,$$

with $a_0, \dots, a_n$ chosen so that $H = 0$ holds *to order* n at x_0 .

Quant_Macro Lec. 9, "Solving Functional Equations". Block A's exposition follows Fernández-Villaverde & Guerrón-Quintana, *Solution Methods* V–VI, rewritten here.

The shift

The unknown stops being *a table of values* and becomes *a short list of derivatives at one point*.

The known point is the **deterministic steady state**: computed since 1.B1 in one line, used so far only as a sanity check. It is about to become the whole solution.

The Toolkit: Five Things, and We Have Four of Them

1. **Implicit function theorem** — licenses writing d as a function of (x, θ) near a reference point at all.
2. **Taylor's theorem** — expands that function, and turns “solve for d ” into “solve for its derivatives”.
3. **Differentiating an identity** — if $F(\cdot) \equiv 0$ then every derivative of F is zero. This is where the equations come from.
4. **An eigenvalue condition** — Blanchard–Kahn, which decides whether the linear system we end up with has one solution, many, or none. *New in 5.A4.*
5. **Complementarity reformulation** — Fischer–Burmeister, when the model has inequalities. *2.A1 and 4.A2.*

Only item 4 is genuinely new. Items 1–3 are calculus; item 5 you have already used twice.

Everything here is **local**. It describes the economy for small deviations from the expansion point and makes no claim beyond them.

The Roadmap, Once, So the Next Three Decks Have a Shape

Step		Where
1	Write the equilibrium conditions as one system $F(\cdot; \chi) = 0$	5.A2
2	Choose the expansion point: deterministic steady state, $z = 0, \chi = 0$	5.A2
3	Replace the policy functions by Taylor series around it	5.A2
4	Differentiate $F = 0$ there; each derivative is an equation	5.A3
5	Solve the linear systems — first order, then higher orders	5.A3, 5.A5
6	Check existence and uniqueness, and ask how far it is valid	5.A4, 5.A6

Notice what is missing

No grid, no fixed point, no maximization. Steps 4 and 5 are differentiation and linear algebra — which is the entire reason it is fast.

Quant_Macro Lec. 9, "Roadmap: Solving a DSGE Model by Perturbation". χ is the shock-scale parameter of 5.A2.

The Implicit Function Theorem: Permission to Start

Before expanding d in a series we need to know that a function d exists at all.

IFT. If $H(d_0, x_0, \theta_0) = 0$ and

$$\frac{\partial H}{\partial d}(d_0, x_0, \theta_0) \neq 0,$$

then on a neighbourhood of (x_0, θ_0) there is a unique

$$d = f(x, \theta)$$

solving $H = 0$, and it is as smooth as H is.

So the non-singularity of one Jacobian buys three things at once: **existence**, **local uniqueness**, and **differentiability** — and the third is what we are about to spend.

Quant_Macro Lec. 9, "Role of the Implicit Function Theorem".

Local, and only local

The theorem says nothing outside the neighbourhood, and never says how large it is. Every limitation of perturbation is already visible in this sentence.

Compare 3.A1: there, "solve" meant a contraction mapping with a global fixed point. Here it means a local inverse. Two different senses of the same word.

Two Derivatives That Look Like a Contradiction

Students get stuck here, so it is worth being explicit. Two derivatives appear, and they point opposite ways.

One must be non-zero:

$$\frac{\partial H}{\partial d} \neq 0 \quad \text{at } (d_0, x_0, \theta_0).$$

This is solvability *in the d direction* — the IFT hypothesis.

The other must be zero. Once $d = f(x, \theta)$ is the solution, define

$$G(x, \theta) := H(f(x, \theta), x, \theta) \equiv 0,$$

identically, on the whole neighbourhood. So

$$\frac{\partial G}{\partial x} = \frac{\partial G}{\partial \theta} = 0.$$

Quant_Macro Lec. 9, "Why the Derivatives Matter".

No contradiction

They are derivatives of *different functions*. H is a map with d as a free argument; G is what is left after the solution has been substituted in — and it is the zero function.

The first derivative is a *hypothesis*. The second is the *source of every equation* we are going to write down.

Is the Hypothesis Satisfied? In the Growth Model, Yes

Here H is the pair (Euler equation, resource constraint) and $d = (c, k')$. The relevant Jacobian is

$$\frac{\partial H}{\partial (c, k')} = \begin{pmatrix} \partial_c(\text{Euler}) & \partial_{k'}(\text{Euler}) \\ \partial_c(\text{budget}) & \partial_{k'}(\text{budget}) \end{pmatrix}.$$

It has full rank at the steady state for two economic reasons:

- $u''(c) < 0$: concavity makes the Euler equation respond to c at all.
- The budget constraint is **linear** in (c, k') , coefficient 1 on each.

Were it singular, the model would be degenerate — one more unit of saving would cost nothing.

Quant_Macro Lec. 9, "How Is $\partial H / \partial d \neq 0$ Satisfied in Growth Models?".

Where it returns

Blanchard–Kahn (5.A4) checks the eigenvalues of the *linearized* system. Passing it confirms, after the fact, that this Jacobian was invertible — the same condition from the other end.

$u'' < 0$ has now done four jobs: uniqueness of the maximum (1.B2), unimodality for golden section (4.A2), the constant in Santos's bound (3.A1), and now local solvability.

The Chain Rule Is the Whole Method

With $G(x, \theta) = H(f(x, \theta), x, \theta) \equiv 0$, differentiate:

$$\frac{\partial G}{\partial x} = \underbrace{\frac{\partial H}{\partial d}}_{\text{known}} \underbrace{\frac{\partial f}{\partial x}}_{\text{unknown}} + \underbrace{\frac{\partial H}{\partial x}}_{\text{known}} = 0,$$

$$\frac{\partial G}{\partial \theta} = \frac{\partial H}{\partial d} \frac{\partial f}{\partial \theta} + \frac{\partial H}{\partial \theta} = 0.$$

Everything except $\partial f / \partial x$ can be evaluated at the steady state, because we know the steady state. So this is a **linear equation in the unknown derivative**, and the IFT hypothesis is exactly what makes it solvable.

Repeat at second order and you get a linear system for the second derivatives, in which the first derivatives now appear as *known* coefficients. The procedure is recursive.

Quant_Macro Lec. 9, "Chain Rule and Matching Conditions".

Match coefficients, order by order.

Each order is linear given the ones below it — which is why fifth-order perturbation is not appreciably harder than second, only messier.

The first order is the only one that requires a decision (5.A4). Everything above it is bookkeeping.

What If the Model Has Inequalities?

The method above needs a system of **equalities**, $F(\cdot) = 0$. Models with a borrowing limit or irreversible investment give Kuhn–Tucker conditions instead:

$$a \geq 0, \quad h \geq 0, \quad ah = 0.$$

The **Fischer–Burmeister** function turns the triple into one equation:

$$\Psi^{FB}(a, h) = a + h - \sqrt{a^2 + h^2} = 0.$$

Check the three cases: $(a > 0, h = 0)$ gives $a - a = 0$; $(a = 0, h > 0)$ gives $h - h = 0$; $(0, 0)$ gives 0. And if both are strictly positive, $\Psi^{FB} < 0$ by Cauchy–Schwarz — so it is zero exactly on the complementarity set.

Quant_Macro Lec. 9, “Perturbation Methods” (the complementarity frame); same notation as 2.A1 and 4.A2.

But be honest about it

Ψ^{FB} is **not** differentiable at the origin — precisely where the constraint starts to bind. Perturbation needs derivatives. Reformulating the kink does not remove it.

This is why Block B exists, and why the Aiyagari economy of 2.A cannot be perturbed at the constraint. We come back to it in 5.A6.

Takeaways

- Perturbation changes the **representation** of the unknown: from N values on a grid to a handful of **derivatives at one point**.
- The implicit function theorem is what licenses it, and everything local about the method is already contained in that theorem's fine print.
- Two derivatives, opposite requirements: $\partial H / \partial d \neq 0$ is the *hypothesis*; $\partial G / \partial (x, \theta) = 0$ is the *source of the equations*.
- “Differentiate an identity, match coefficients, solve a linear system, repeat” — and each order is linear given the ones below it.
- Inequality constraints can be written as equalities, but the **kink survives the rewriting**. That limitation is structural, not technical.
- Measured, on Session 4's own model: **four numbers** against 2,100, and better accuracy near the steady state — but no better far from it.

Bridge: A Model, and One Extra Parameter

So far this has been calculus with no economics in it. The next deck puts a model underneath: the stochastic growth model with log utility and full depreciation — the Brock–Mirman benchmark this course has used since 1.B1, and the one model whose policy we know exactly.

It also introduces the trick that makes uncertainty tractable: an artificial parameter that **scales the shock**, so that “no risk” is a point we can expand around.

Next (5.A2)

The model, its closed-form solution, the steady state, and the perturbation parameter χ : why we expand around $\chi = 0$ rather than around the model we actually want.

Homework 1 is due today. The practice midterm goes out this afternoon; the midterm itself is next week, Block B, and covers Sessions 1–5.